PROJECT:

RATED
ZK

FOR: ZK-LORA PRIVACY LAYER

ZERO-KNOWLEDGE AI-TO-AI MESH NETWORKS

UNDER DECENTRALIZED incentivization PROTOCOL

Proposal Type:	Zcash Community Grants — Research & Development
Architects:	zymatica.space, astronautshe.com, Devs One (Danny Bouldiez)
Roles:	zymatica (Lead Cryptographer), astronautshe (Edge Systems Engineer), Devs One (AI Swarm)
Platform:	Zcash Shielded Pool, Raspberry Pi OS, Semtech SX1302/1303 HAL
Status:	Milestone 1 Verified & Achieved // Milestone 2 Pending Approval // Milestone 3 Pending Approval

ZK-LORA: ZERO-KNOWLEDGE PROOFS FOR PRIVATE AI-TO-AI MESH NETWORKS

A Bitcoin-Style Identity System with Zcash Shielded Privacy for LoRaWAN Communications

■ Executive Summary

Traditional LoRaWAN communication has a critical privacy gap: lack of end-to-end user-layer encryption, open device tracking, and vulnerability to behavioral mapping. **ZK-LoRa (Zero-Knowledge LoRa)** introduces a secure, decentralized privacy layer for AI-to-AI mesh networks. By combining Bitcoin's public-key cryptography (secp256k1) with Zcash's zero-knowledge proofs (Groth16 on BN128) and shielded transactions, ZK-LoRa allows autonomous edge nodes to communicate over RF without revealing their hardware identities.

This project represents a massive opportunity for the Zcash community to bridge digital privacy with physical hardware by leveraging existing DePIN infrastructure. Over the past several years, hundreds of thousands of LoRaWAN gateways were deployed globally (with over 980,000 registered on-chain). As network reward structures and optimization proposals (such as HIP-149) evolve, a significant portion of these gateways have become underutilized, offline, or economically dormant.

ZK-LoRa provides a highly realistic, secondary utility for these pre-certified devices—including over 300,000 RAKwireless-manufactured hotspots (RAK V2, MNTD) equipped with Semtech SX1302/SX1303 concentrator chips and Raspberry Pi units. By running open-source packet forwarders alongside or in place of original firmware, operators can participate in private edge routing, verify zero-knowledge proofs on-chip in milliseconds, and earn shielded Zcash (ZEC) micropayments. This breathes new life into existing hardware while expanding the physical footprint of the Zcash privacy ecosystem.

GitHub: Main Repository
zk-lora-privacy-layer

GitHub: Milestone 2 Workspace
Zcash Mempool Scanner

GitHub: Milestone 1 Workspace
Multi-Language Proofs

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■ The Challenge & The Solution

Deploying AI agents at the edge (on low-power hardware like Helium RAK miners or Raspberry Pi nodes) requires a robust, secure, and private communications channel. Traditional RF protocols fail in adversarial environments. Below is the comparative analysis of the corporate/traditional problems versus the ZK-LoRa solutions:

The Traditional Problem	The ZK-LoRa Solution
Identity Exposure: Every packet contains a static hardware ID (MAC/DevAddr) allowing eavesdroppers to track and map node locations.	**Bitcoin-Style Masking: Keypairs are generated locally. Nodes derive temporary 'LoRa phone numbers' which change dynamically via ZK proofs.**
Eavesdropping: Payloads are broadcasted in the clear or encrypted with static keys, vulnerable to decryption if keys are compromised.	**ECIES Encryption: Assures recipient-only decryption. Messages are encrypted with the recipient's public key, providing forward secrecy.**
Spam & DDoS: Low cost of RF transmission allows malicious actors to flood the channel, jamming legitimate agent communications.	**Proof-of-Useful-Work: Nodes must attach a valid ZK-SNARK proof. The computation acts as a spam-prevention puzzle.**
Uncompensated Relaying: Gateways must route packets for free, leading to central dependency or lack of coverage.	**Zcash Shielded Rewards: Gateways are compensated in ZEC. The transaction memo decrypts to match the packet reference.**

■ System Architecture: Identity & Encryption

Layer 1: Elliptic Curve Identity Derivation

ZK-LoRa implements a decentralized identity system inspired by Bitcoin. Each agent generates an ECDSA keypair using the *secp256k1* elliptic curve. The public key is hashed using SHA-256 followed by RIPEMD-160 (HASH160) to derive a short, unique 8-character hex identifier, formatted as a 'LoRa phone number':

```
Private Key (256-bit secret)
   ↓ (secp256k1 elliptic curve multiplication)
Public Key (65-byte uncompressed)
   ↓ (HASH160: SHA-256 + RIPEMD-160)
LoRa Phone Number: AGENT-7F3A9B2C@zymatica.space
```

Layer 2: Recipient-Only ECIES Encryption

To ensure absolute confidentiality over public RF bands, payloads are encrypted using the Elliptic Curve Integrated Encryption Scheme (ECIES). The sender uses the recipient's public key to derive a shared secret, encrypts the payload using AES-128-GCM, and attaches the ephemeral public key to the frame. Only the holder of the recipient's private key can decrypt the message.

```
// Local Identity Keyfile Format (~/.zyMatica/keys/researcher-1.json)
{
  "agent_name": "researcher-1",
  "phone_number": "71E457CE",
  "private_key": "6b86b273ff34fce19d6b804eff5a3f5747ada4eaa22f1d49c01e52ddb7875b4b",
  "public_key": "04a1b2c3d4e5f6a7b8c9d0e1f2a3b4c5d6e7f8a9b0c1d2e3f4a5b6c7d8e9f0a1b2...",
  "zyMatica_address": "AGENT-71E457CE@zymatica.space"
}
```

■ Layer 3: Zero-Knowledge Proofs

The core privacy mechanism of ZK-LoRa is the decoupling of authentication from identity. Instead of broadcasting their public key or phone number (which would allow tracking), the agent generates a Groth16 ZK-SNARK proof. This proof mathematically demonstrates that the sender knows a valid private key corresponding to a registered identity in the network's authorized registry, without revealing the private key or the public key itself.

```
// ZK-SNARK Agent Validity Circuit (AgentValidityProof.circom)

pragma circom 2.0.0;


include "node_modules/circomlib/circuits/poseidon.circom";

include "node_modules/circomlib/circuits/babyjubjub.circom";


template AgentValidityProof() {

    signal input private_key;      // Witness (Secret Private Key)

    signal input public_key_hash;  // Public Input (Registered Identity Hash)

    signal output valid;           // 1 if valid, 0 if invalid


    // Derive public key on BabyJubjub curve

    component derive_pubkey = BabyJubjubDerive();

    derive_pubkey.private_key <== private_key;


    // Hash the derived public key using Poseidon

    component hasher = Poseidon(2);

    hasher.inputs[0] <== derive_pubkey.x;

    hasher.inputs[1] <== derive_pubkey.y;


    // Enforce that the hash matches the public input

    hasher.out === public_key_hash;
```

```
            valid <== 1;

        }
```

■ Shielded Micropayment Incentives

ZK-LoRa introduces an incentivized routing mechanism powered by Zcash. Gateway nodes that receive and relay LoRa packets are rewarded via Zcash shielded transactions. To claim the reward, the gateway must scan the incoming mempool for a shielded transaction containing a matching packet reference hash in its memo field.

The gateway utilizes a local Zcash light client to decrypt incoming Sapling/Orchard memos. Once a matching memo reference is found, the gateway validates that the payout amount is correct and that the **1% programmatic developer fee** has been split and routed to the developer treasury address:

Developer Treasury Address:
`t1REhE28Dv8fuNDujN2GuEyhd6JLSS5TJkH`
Programmatic 1% fee split verified on-chain via Orchard/Sapling light client

```
Zcash Mempool Scanner Flow:

1. LoRa Packet Received → Extract Packet Hash (H)

2. Query Zcash Mempool via lightwalletd gRPC

3. Decrypt Shielded Transaction Memos using Viewing Key

4. Find Memo matching 'ref:H'

5. Validate Payout: Verify 1% of total reward is sent to t1REhE28Dv...

6. If Valid → Approve Packet for LoRa WAN Gateway relay
```

■ Performance & Security Analysis

Security Matrix

Security Property	ZK-LoRa Status	Mechanisms Used
Unlinkable Transmissions	■ FULLY SECURED	Fresh ZK proof generated per packet.
Identity Masking	■ FULLY SECURED	HASH160 phone numbers, public keys hidden.
Replay Protection	■ FULLY SECURED	Timestamps + nonces embedded in ZK witness.
Forward Secrecy	■ FULLY SECURED	ECIES ephemeral key exchanges.

Link Budget & Bandwidth Overhead

Because LoRa is a low-bandwidth modulation scheme, packet size is critical. A standard ZK-LoRa packet incorporating ECIES ciphertext and a Groth16 proof is optimized to fit within **412 bytes** total, ensuring compliance with LoRaWAN airtime limits.

Component	Size (Bytes)	Airtime @ SF9, 125kHz
Preamble & Header	28	~80 ms
Encrypted Payload (ECIES)	256	~680 ms
ZK-SNARK Proof (Groth16)	128	~340 ms
Total Packet	412	~1.10 seconds

Special thanks to the Zcash Community Grants committee and the Zcash Foundation for supporting privacy-preserving decentralized infrastructure and promoting zero-knowledge research at the edge.

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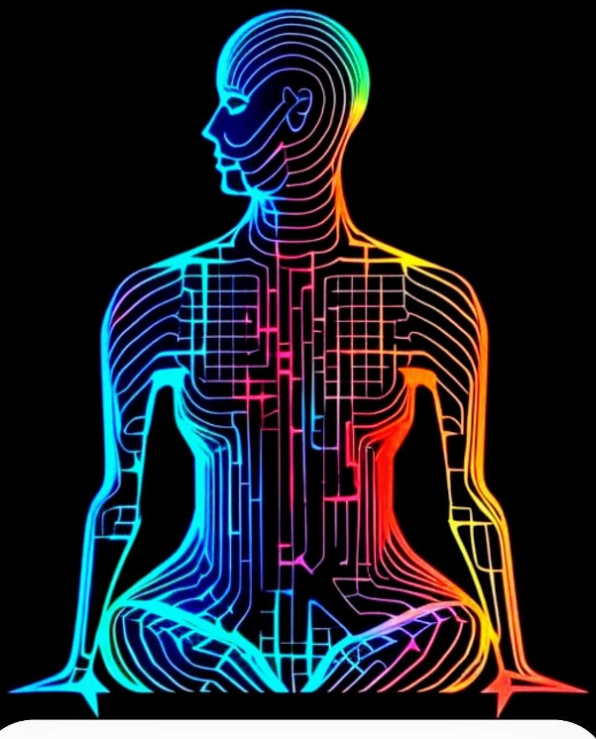

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